This ensures that auxiliary-variable connectivity emerges naturally from the learned rule structure.

Digitization Loss (λ_D). The digitization loss encourages integer-valued variable usage counts. Because this loss is highly non-convex, it is introduced cautiously using:

$$\lambda_D(t) = \rho^2 \, \lambda_D^{\max}.$$

In practice, $\lambda_D^{\max}$ is kept small to avoid destabilizing training.

Final Objective:

With scheduling applied, the total loss at epoch t becomes:

$$\mathcal{L}(t) = \mathcal{L}_{\text{BCE}} + \lambda_E(t)\mathcal{L}_E + \lambda_S(t)\mathcal{L}_S + \lambda_R(t)\mathcal{L}_R + \lambda_C(t)\mathcal{L}_C + \lambda_D(t)\mathcal{L}_D.$$

This scheduling strategy was used consistently across all experiments unless otherwise stated. Empirically, it significantly improved optimization stability, reduced premature rule collapse, and enabled reliable symbolic rule extraction—particularly under probabilistic inputs and label noise.

E PREDICATE IDENTIFICATION

To ensure interpretability, it is essential to explicitly extract symbolic subrules after training. Although the model learns each probability distribution P_{ij} for $\mathtt{B}_{ij}$, which reflects the importance of candidate subpredicates, a post-processing step is required to derive the final symbolic rules.

During training, ANDRE encourages one of the probabilities in P_{ij} to converge to 1, with the remaining values approaching 0. In this case, the corresponding symbolic subpredicate is considered the most confident selection: if $p_{ij}^{(1)} \to 1$, $p_{ij}^{(2)} \to 1$, or $p_{ij}^{(3)} \to 1$, then $\mathtt{b}_j$, $\neg\mathtt{b}_j$, or $\mathtt{1}$ is selected, respectively.

In cases where none of the probabilities fully converges to 1, but one clearly dominates, identifying the most confident subpredicate requires evaluating the certainty of the distribution. To do so, the entropy of the distribution is computed as: $\mathcal{H} = \sum_{k=1}^{3} p_{ij}^{(k)} \log\left(p_{ij}^{(k)} + \epsilon\right)$, where ϵ is a small constant added for numerical stability. Lower entropy values indicate greater certainty. Based on this entropy, the most confident subpredicate $\mathtt{B}_{ij}^{\text{andre}}$ is extracted using the following criterion:

$$\mathtt{B}_{ij}^{\text{andre}} = \begin{cases} \mathtt{S}_j\left[\arg\max_k \, \mathcal{S}_j[k]\right], & \text{if } \mathcal{H} \leq \eta' \\ \mathtt{1}, & \text{otherwise}, \end{cases} \tag{18}$$

where $\mathtt{B}_{ij}^{\text{andre}} \in \mathtt{B}^{n\times m}$ denotes the extracted symbolic subpredicate, and n and m are the number of subrules and body predicates, respectively. The constant $\eta' \in [0, \ln(3)]$ is selected based on the model's final accuracy and the noise level in the data. A lower value of η' imposes a stricter certainty requirement for subpredicate selection.

After this selection process, ANDRE constructs the matrix $\mathtt{B}_{ij}^{\text{andre}}$, where each entry corresponds to one of the symbolic subpredicates: $\mathtt{b}_j$, $\neg\mathtt{b}_j$, or $\mathtt{1}$. These entries then replace $\mathtt{B}_{ij}$ in Eq. (4) to form the subrules, thereby producing the final rule as specified in Eq. (2).

F ANDRE'S PARAMETERS

Table 5 summarizes the hyperparameters used in all experiments with ANDRE's framework. These parameters were selected based on empirical tuning and prior best practices in differentiable rule learning. They govern various aspects of optimization, loss weighting, attention sharpness, and symbolic rule extraction behavior. Where applicable, different values are recommended under noisy conditions to improve robustness and generalization.

Table 5: Hyperparameter Settings Used in ANDRE

Parameter	Symbol	Value
Learning rate	α	0.01
Learning rate decay	α'	0.0001
Sigmoid center	γ	0.5
Range-Restricted Loss Coefficient	η	0.1
Connected loss coefficient	c_1	1.0
Connected loss coefficient	c_2	12.5
Attention sharpness coefficient	β	20
Sigmoid sharpening factor	λ	10
Number of random restarts per subrule	N_r	3
Training epochs per restart	T	500
Batch size	b	$[128, 512, 4096]$
Similarity loss coefficient	λ_{S}	0.2
Entropy loss coefficient	λ_{E}	0.1
Range-restricted loss coefficient	λ_{R}	1.00
Connected loss coefficient	λ_{C}	5.00
Digitization loss coefficient	λ_{D}	0.001
Accuracy threshold for early stopping	τ	0.95 (reduced in noisy settings)
Entropy threshold for predicate certainty	η'	0.4 (standard), 0.8 (noisy data)

G Visual Examples of ANDRE

G.1 Structured Rule: Grandparent Task

To further illustrate the internal learning dynamics and interpretability of ANDRE on structured relational data, we present a visual case study on the *Grandparent* task. This task involves learning a first-order rule with auxiliary variables and multiple valid subrule instantiations, making it a representative benchmark for evaluating both semantic and syntactic learning behavior.

Task Description. The objective is to learn the head predicate $\texttt{grandparent}(X_1, X_3)$, given background knowledge consisting of `mother` and `father` relations. The correct logical definition requires identifying transitive parent relationships through an auxiliary variable X_2:

$$\texttt{grandparent}(X_1, X_3) \leftarrow \texttt{parent}(X_1, X_2) \wedge \texttt{parent}(X_2, X_3),$$

where `parent` may be instantiated as either `mother` or `father`.

Instead of the propositionalization strategy, the trainable samples are generated using random probabilistic predicate valuations and the known target logical rules, resulting in a supervised dataset $\mathcal{E} \in \{0,1\}^{N\times(m+1)}$, where each input example consists of valuations of candidate body predicates

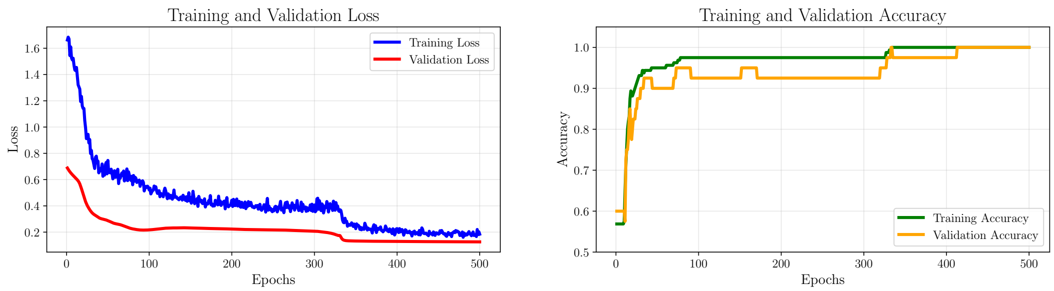
$$\mathcal{B} = \{\texttt{mother}(X_1, X_2), \texttt{mother}(X_2, X_3), \texttt{father}(X_1, X_2), \texttt{father}(X_2, X_3), \dots\},$$

and a Boolean label indicating whether the corresponding `grandparent` fact is true. In this example, we generate 1000 samples, and $m = 6$ and $N = 4$ to better visualize the results. ANDRE is trained end-to-end without prior knowledge of rule templates, explicit clause enumeration, or predicate relevance.

Optimization Dynamics. During training, ANDRE jointly optimizes semantic and syntactic objectives using the loss formulation in Eq. (17). The binary cross-entropy loss guides semantic correctness, while entropy, range-restricted, connectedness, digitization, and similarity losses gradually enforce rule sparsity, variable grounding, and structural validity. Loss coefficients are scheduled over training as described in Appendix D to stabilize optimization and avoid premature structural constraints.

Figure 6 shows the evolution of the total loss during training. After an initial phase dominated by semantic alignment, the loss decreases steadily as the model converges to a stable and interpretable rule structure.

Learned Predicate Weights and Rule Extraction. Upon convergence, ANDRE produces sharp probability distributions over the symbolic subpredicate space $\{b_j, \neg b_j, 1\}$ for each body predicate



(a) Loss (b) Accuracy

Figure 6: Optimization behavior of ANDRE on the Grandparent task. Left: evolution of training and validation loss. Right: corresponding training and validation accuracy. The curves indicate stable convergence and strong generalization.

in each subrule. High-confidence selections correspond to predicates that consistently participate in valid grandparent relationships, while irrelevant predicates are suppressed.

Figure 4 illustrates the final learned subpredicate weights. Accordingly, after applying the predicate identification procedure described in Appendix E, ANDRE extracts the following symbolic rule set for the `grandparent` task:

$$\begin{aligned}
&\texttt{grandparent}(X_1, X_3) \leftarrow \texttt{mother}(X_1, X_2) \wedge \texttt{mother}(X_2, X_3),\\
&\texttt{grandparent}(X_1, X_3) \leftarrow \texttt{father}(X_1, X_2) \wedge \texttt{father}(X_2, X_3),\\
&\texttt{grandparent}(X_1, X_3) \leftarrow \texttt{mother}(X_1, X_2) \wedge \texttt{father}(X_2, X_3),\\
&\texttt{grandparent}(X_1, X_3) \leftarrow \texttt{father}(X_1, X_2) \wedge \texttt{mother}(X_2, X_3),
\end{aligned}$$

which are both semantically and syntactically correct. Variables X_1, X_3 are head variables, while X_2 is an auxiliary variable.

For each rule, we present the accuracy over training and validation sets. Also, we consider $(N_b, N_r, \frac{N_r}{N_b})$ as an evaluation metric showing the quality of each rule. N_b is the number of trainable samples each rule body satisfies, while N_r is the number of trainable samples that both rule body and head satisfy. When $\frac{N_r}{N_b}$ is higher, it means the extracted rule is satisfying all relevant samples in the training samples. Below are the extracted rules for the grandparent task:

Listing 1: Extracted Rules for $\texttt{grandparent}(X_1, X_3)$ (Synthetic Dataset)

```
[1] grandparent(X1, X3) :- father(X1, X2) and mother(X2, X3).
  Training Acc: 0.6825 | Eval Acc: 0.6900
  Val Coverage: (N_b=53, N_r=53, N_r/N_b=1.00)
  Train Coverage: (N_b=219, N_r=219, N_r/N_b=1.00)

[2] grandparent(X1, X3) :- father(X1, X2) and father(X2, X3).
  Training Acc: 0.6725 | Eval Acc: 0.6850
  Val Coverage: (N_b=52, N_r=52, N_r/N_b=1.00)
  Train Coverage: (N_b=211, N_r=211, N_r/N_b=1.00)

[3] grandparent(X1, X3) :- mother(X1, X2) and mother(X2, X3).
  Training Acc: 0.6850 | Eval Acc: 0.6500
  Val Coverage: (N_b=49, N_r=47, N_r/N_b=0.9592)
  Train Coverage: (N_b=225, N_r=223, N_r/N_b=0.9911)

[4] grandparent(X1, X3) :- father(X2, X3) and mother(X1, X2).
  Training Acc: 0.6538 | Eval Acc: 0.65
  Val Coverage: (N_b=45, N_r=45, N_r/N_b=1.00)
  Train Coverage: (N_b=196, N_r=196, N_r/N_b=1.00)
```

These subrules collectively form a complete and interpretable definition of the grandparent relation, fully consistent with first-order logic and human intuition.

Discussion. This visual example demonstrates ANDRE's ability to learn multi-clause first-order rules directly from data using a fully differentiable optimization process. The model correctly han-